Tests and Indices in Predicting Extubation Failure in Children: A Systematic Review and Meta-analysis

SUPPLEMENTARY MATERIAL

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Table S1 Search strategy for respective databases

Database	Search strategy / Keywords
PubMed	#1 – "Airway Extubation"[Mesh] #2 – "Ventilator Weaning"[Mesh] #3 – Extubat* OR "ventilator weaning" #4 – #1 OR #2 OR #3 #5 – #4 AND Filter Ages (Child: Birth – 18 years old)
Embase	#1 – 'extubation'/exp #2 – 'ventilator weaning'/exp #3 – Extubat* OR "ventilator weaning" #4 – #1 OR #2 OR #3 #5 – #4 AND Filter Ages (Newborn, Infant, Child (1-12), Preschool child (1-6), School child (7-12), Adolescent)
Web of Science	#1 – TS=(exubat*) OR TI=(extubat*) OR SU=(extubat*) #2 – TS=(ventilator weaning) OR TI=(ventilator weaning) OR SU=(ventilator weaning) #3 – #1 OR #2 #4 – TS=(child*) OR TI=(child*) OR SU=(child*) #5 – TS=(adolescen*) OR TI=(adolescen*) OR SU=(adolescen*) #6 – TS=(pediatric*) OR TI=(pediatric*) OR SU=(pediatric*) #7 – TS=(paediatric*) OR TI=(paediatric*) OR SU=(paediatric*) #8 – #7 OR #6 OR #5 OR #4 #9 – #8 AND #3

CINAHL	S1 – (MH “Extubation”) S2 – (MH “Ventilator weaning”) S3 – “ventilator weaning” OR “extubat*” S4 – S1 OR S2 OR S3 S5 – S4 AND Filter ages (Child, preschool: 2-5 years; adolescent: 13-18 years; child: 6-12 years; infant: 1-23 months; infant, newborn: birth – 1 month; all infant; all child)
Cochrane	#1 - MeSH descriptor: [Airway Extubation] explode all trees #2 - MeSH descriptor: [Ventilator Weaning] explode all trees #3 - extubat* OR ventilator weaning #4 - #1 OR #2 OR #3 #5 - MeSH descriptor: [Child] explode all trees #6 - MeSH descriptor: [Pediatrics] explode all trees #7 - MeSH descriptor: [Adolescent] explode all trees #8 – paediatric* OR pediatric* OR child* OR adolescen* #9 - #5 OR #6 OR #7 OR #8 #10 - #4 AND #9

Table S2 Studies on bedside vital signs

Bedside vital signs	Number of studies	Total number of patients in included studies	Studies that studied difference in vital signs between extubation failure and extubation success group	Studies that studied association of vital sign with EF (odds ratio)	Studies that studied diagnostic accuracy of vital signs predicting EF
RR	13 (1–13)	1945	13 (1–13)	1 (12)	5 (1, 3–5)
SpO_2	4 (6, 10–12)	815	4 (6, 10–12)	1 (12)	-
HR	1 (6)	409	1 (6)	-	-
$EtCO_2$	1 (10)	45	1 (10)	-	-
Cumulative fluid balance	1 (14)	301	1 (14)	-	-

** Shortlisted for diagnostic meta-analysis*

RR = Respiratory rate, SpO2 = oxygen saturation, HR = Heart rate, $EtCO_2$ = End-tidal CO_2

Table S3 Studies on blood tests

Blood test	Number of studies	Total number of patients in included studies	Studies that studied difference in blood test result between EF and extubation success group	Studies that studied association of blood test with EF (odds ratio)	Studies that studied diagnostic accuracy of blood test predicting EF
$PaCO_2$	10 (1, 3–5, 7, 10–13, 15)	1379	9 (1, 3–5, 7, 10, 11, 15, 13)	1 (12)	1 (3)
PaO_2	9 (1, 4, 5, 7, 8, 10, 12, 13, 15)	1460	8 (1, 4, 5, 7, 8, 10, 13, 15)	1 (12)	-
Arterial pH	5 (4, 10–13)	673	4 (4, 10, 11, 13)	1 (12)	-
HCO_3^-	2 (10, 11)	87	2 (10, 11)	-	-
Venous oxygen saturation	1 (12)	319	-	1 (12)	-
Venous pH	1 (12)	319	-	1 (12)	-
Venous pCO_2	1 (12)	319	-	1 (12)	-
Venous pO_2	1 (12)	319	-	1 (12)	-
Base excess	1 (10)	45	1 (10)	-	-

** Shortlisted for diagnostic meta-analysis*

PaCO2 = Partial pressure of arterial carbon dioxide; PaO2 = Partial pressure of arterial oxygen; HCO_3^- =Bicarbonate; pCO_2 = partial pressure of carbon dioxide; pO_2 = partial pressure of oxygen

Table S4 Studies on ventilator parameters

Ventilator parameter	Number of studies	Total number of patients in included studies	Studies that studied difference in ventilator parameter between EF and extubation success group	Studies that studied association of ventilator parameter with EF (odds ratio)	Studies that studied diagnostic accuracy of ventilator parameter predicting EF
Tidal volume	13 (1–13)	1945	11 (1, 2, 4–11, 13)	2 (12, 13)	6 (1–5, 13)
Pi_{max}	8 (1, 2, 4–7, 13, 16)	1600	8 (1, 2, 4–7, 13, 16)	1(6)	5 (1, 5, 13, 16)
FiO_2	8 (4, 5, 7–10, 12, 13)	954	7 (4, 5, 7–10, 13)	1 (12)	1 (5)
PIP	7 (4, 5, 7, 8, 10, 12, 13)	918	6 (4, 5, 7, 8, 10, 13)	1 (12)	1 (5)
PEEP	6 (4, 6, 10–13)	1082	5 (4, 6, 10, 11, 13)	2 (6, 12)	-
Minute volume	5 (2–4, 8, 13)	539	4 (2, 4, 8, 13)	-	2 (3, 13)
Mean airway pressure	5 (4, 5, 10, 13, 15)	516	5 (4, 5, 10, 13, 15)	1 (15)	1 (5)
Compliance	4 (5, 6, 10, 11)	576	4 (5, 6, 10, 11)	-	1 (5)
Pressure Support	3 (10–12)	406	2 (10, 11)	1 (12)	-
Inspiratory time	3 (5, 6, 13)	529	3 (5, 6, 13)	-	1 (5)
Airway resistance	3 (5, 10, 11)	167	3 (5, 10, 11)	-	1 (5)
T_{TOT}	2 (5, 6)	489	2 (5, 6)	-	1 (5)
Dynamic compliance	2 (3, 4)	274	1 (4)	-	1 (3)
Inspiratory pressure	1 (5)	80	1 (5)	-	1 (5)
Plateau pressure	1(11)	42	1(11)	-	-
Maximum esophageal pressure	1(6)	409	1(6)	-	-

FRC	1(5)	80	1(5)	-	1(5)
Phase angle	1 (6)	409	1 (6)	1 (6)	-
Crying vital capacity	1 (16)	294	-	-	1(16)
P_{breath}	1 (3)	47	-	-	1 (3)
ROP	1 (7)	57	-	-	1 (7)
$P_{0.1}$	1 (5)	80	1 (5)	-	1 (5)

** Shortlisted for diagnostic meta-analysis*

Pi_{max} = Maximal inspiratory pressure; FiO_2 = Fraction of inspired oxygen; PIP = Peak inspiratory pressure; PEEP = Positive end-expiratory pressure; T_{TOT} = Total time in one breath; FRC = Functional residual capacity; P_{breath} = the product of the measured pressure change during a mechanical inflation (peak inspiratory pressure - positive end-expiratory pressure) and the ratio of the spontaneous tidal volume and the volume delivered by a positive pressure inflation; $P_{0.1}$ = Airway occlusion pressure

Table S5 Studies on collective pre-extubation assessments

Index	Number of studies	Total number of patients in included studies	Studies that studied difference in index between EF and extubation success group	Studies that studied association of index with EF (odds ratio/relative risk)	Studies that studied diagnostic accuracy of index predicting EF
RSBI*	9 (1–7, 13, 17)	1400	6 (1, 2, 4–6, 13)	1 (4)	8 (1–5, 13, 17)
CROP index*	4 (3–5, 17)	401	2(4, 5)	-	4(3–5, 17)
PaO_2/FiO_2 ratio	4 (1, 2, 4, 7)	777	7 (1, 2, 4, 7)	-	1 (1)
Pi/Pi_{max}	4 (4–6, 18)	766	4 (4–6, 18)	-	2(5, 18)
Tension time index	3 (5–7)	546	2 (5, 6)	-	2 (5, 7)
T_i/T_{TOT} ratio	3 (3, 5, 6)	536	2 (5, 6)	-	2 (3, 5)
Vd/Vt	3 (10, 11, 19)	87	2 (10, 11, 19)	1 (19)	2 (10, 11)
A-a ratio	2 (2, 3)	122	1 (2)	-	1 (3)
Oxygenation index	2 (13, 15)	164	2 (13, 15)	1 (15)	-
Pressure rate product	1 (6)	409	1 (6)	-	-
Pressure time product	1 (6)	409	1 (6)	-	-
Pressure time index	1 (7)	57	-	-	1 (7)
Pdi/Pdi_{max}	1 (5)	80	1(5)	-	1(5)
Load/force balance	1 (13)	40	1(13)	1(13)	1(13)
Combination of index (permutations involving RR, Pi_{max}, RSBI, ROP)	1(7)	57	-	-	1(7)

** Shortlisted for diagnostic meta-analysis*

RSBI = Rapid Shallow Breathing Index; CROP index = Compliance rate oxygenation pressure index; PaO_2/FiO_2 ratio = Partial pressure of arterial oxygen/Fraction of inspired oxygen ratio; Pi/Pi_{max} = inspiratory pressure/maximal inspiratory pressure ratio, $Ti/TTOT$ ratio = Inspiratory time / Total time in one breath ratio, Vd/Vt = dead space, A-a ratio = Alveolar-arterial ratio, Pdi/Pdi_{max} = Transdiaphragmatic pressure / maximal transdiaphragmatic inspiratory pressure ratio, RR = Respiratory rate, Pi_{max} = Maximal inspiratory pressure, ROP = Rapid shallow breathing occlusion pressure

Table S6 Studies on spontaneous breathing trials (SBT)

Test	Number of studies	Total number of patients in included studies	Studies that studied difference in result of assessments between EF and extubation success group (or between intervention and control group)	Studies that studied association of assessments with EF (odds ratio / relative risk)	Studies that studied diagnostic accuracy of assessments predicting EF
SBT*	13 (9, 12, 20–30)	5652	7 (21–25, 27, 28)	4 (12, 26, 29, 30)	5 (9, 20–22, 29)
Weaning protocol + SBT	2 (31, 32)	297	2 (31, 32)	-	-

** Shortlisted for diagnostic meta-analysis*

SBT = Spontaneous breathing trials

Table S7 Other pre-extubation assessments

Test	Number of studies	Total number of patients in included studies	Studies that studied difference in result of assessments between EF and extubation success group (or between intervention and control group)	Studies that studied association of assessments with EF (odds ratio / relative risk)	Studies that studied diagnostic accuracy of assessments predicting EF
Diaphragm ultrasound	6 (33–38)	338	5 (33–35, 37, 38)	-	2 (33, 36)
ETT leak test	3 (39–41)	355	-	2 (39)	2 (39, 40)

Table S8 Risk of Bias assessment result for RCTs – (RoB 2.0)

	Domain 1: Randomisation	Domain 2: Assignment to intervention	Domain 3: Missing outcome data	Domain 4: Outcome measurement	Domain 5: Selection of reported outcome	Overall risk of bias
Bilan 2014	-	X	X	X	-	X
El-beleidy 2013	+	+	+	+	+	+
Farias 2001	+	+	+	X	X	X
Foronda 2011	+	+	+	-	+	-
Keivanfar 2020	-	+	+	-	+	-
Randolph 2002	+	+	+	X	X	X

+ *Low risk of bias;* - *Some concerns;* X *High risk of bias*

Table S9 Risk of bias assessment results for cohort studies

Author, Year	SELECTION				COMPARABILITY	OUTCOME		
	Representative-ness of exposed cohort	Selection of non-exposed cohort	Ascertain-ment of exposure	Demonstration that outcome of interest was not present at the start of study	Does the study design or analysis control for confounders	Assess-ment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow-up of cohorts
Abu-Sultaneh 2017	★	★	★	★	-	★	★	★
Baumeister 1997	★	★	★	★	★	★	★	★
Chavez 2006	-	★	★	★	★★	★	★	★
El-Khatib 1996	★	★	★	★	-	★	★	★
Farias 1998	★	★	★	★	★★	★	★	★
Farias 2002	-	★	★	★	-	★	★	★
Faustino 2017	★	★	★	★	★★	★	★	★
Fontela 2005	★	★	★	★	★★	★	★	★
Gehlbach 2020	★	★	★	★	★★	★	★	★
Harikumar 2009	★	★	★	★	-	★	★	★
Hubble 2000	-	★	★	★	★★	★	★	★
Ijland 2020	-	★	★	★	-	★	★	★
Johnston 2010	-	★	★	★	-	★	★	★
Jouvet 2007	-	★	★	★	-	★	★	★
Khemani 2017	-	★	★	★	★★	★	★	★
Krasinkiewicz 2021	★	★	★	★	★★	★	★	★
Laham 2015	★	★	★	★	★★	★	★	★
Loberger 2020	-	★	★	★	-	★	★	★
Manczur 2000	★	★	★	★	-	★	★	★
Mistri 2020	-	★	★	★	★★	★	★	★
Nascimento 2017	-	★	★	★	★	★	★	★
Noizet 2005	★	★	★	★	★★	★	★	★
Rahman 2020	-	★	★	★	-	★	★	★
Randolph 2005	-	★	★	★	★★	★	★	★
Riou 2012	-	★	★	★	-	★	★	★
Subhash 2021	-	★	★	★	★★	★	★	★
Thiagarajan 1999	★	★	★	★	★★	★	★	★

Valverde Montoro 2021	★	★	★	★	★★	★	★	★
Wratney 2008	★	★	★	★	-	★	★	★
Xue 2020	-	★	★	★	★★	-	-	★

Table S10 Risk of bias assessment results for case-control studies

Author, Year	SELECTION				COMPARABILITY	OUTCOME		
	Is the case definition adequate?	**Representativeness of the Cases**	**Selection of controls**	**Definition of controls**	**Are the cases or control matched in design and/or confounders adjusted for in the analysis**	**Ascertainment of exposure**	**Same method of ascertainment for cases and controls**	**Non-response rates**
Adderley & Mullins 1987	-	-	★	★	-	-	-	★
Ferguson 2011	★	★	★	★	★	★	★	★
Mhanna 2002	★	★	★	★	-	★	★	★
Silva-Cruz 2018	★	★	★	★	★	★	★	★
Toida 2017	★	★	★	-	-	★	★	★

Table S11 Subgroup analyses for Rapid Shallow Breathing Index (RSBI) studies

Characteristics of studies	Studies included	Pooled sensitivity (95% CI)	Pooled specificity (95% CI)	Pooled LR+ (95% CI)	Pooled LR- (95% CI)	Pooled DOR (95% CI)	SROC curve (AUC)
All studies	Baumeister 1997 Farias 1998 Farias 2002b Harikumar 2009 Johnston 2010 Manczur 2000 Thiagarajan 1999	0.44 (0.35 - 0.54) I^2 = 74.6%	0.81 (0.78 - 0.84) I^2 = 94.9%	2.56 (1.62 - 4.08) I^2 = 59.7%	0.74 (0.49 - 1.12) I^2 = 78.5%	3.50 (1.27 - 9.61) I^2 = 72.1%	0.64
Studies using CPAP and defined EF as reintubation in 24 hours	Harikumar 2009 Manczur 2000 Thiagarajan 1999	0.58 (0.41 - 0.75) I^2 = 80%	0.8 (0.75 - 0.84) I^2 = 87.7%	2.98 (2.20 - 4.02) I^2 = 0%	0.59 (0.24 - 1.47) I^2 = 87.2%	6.98 (3.03 - 16.08) I^2 = 0%	0.82

Farias 2002b = results taken from the end of 120 min of SBT

LR+ = Positive likelihood ratio; LR- = Negative likelihood ratio; DOR = Diagnostic odds ratio; SROC = Summary receiver operating characteristic; AUC = Area under curve; CPAP = Continuous positive airway pressure; EF = extubation failure, I^2= Statistical heterogeneity; SBT = Spontaneous breathing trials

Table S12 Description of SBT method for studies included in meta-analysis

Studies included	Method of SBT	Duration of SBT
Chavez 2006	Flow-initiation anaesthesia bag which provides CPAP 5cmH_2O via the leak valve	15 minutes
El-beleidy 2013a	ATC 100%, CPAP 5cmH_2O, FiO_2 <0.5,	2 hours
El-beleidy 2013b	Pressure support adjusted to ETT size, CPAP 5cmH_2O, FiO_2 <0.5	2 hours
Faustino 2017	Pressure support adjusted to ETT size, CPAP 5cmH_2O, FiO_2 0.5	2 hours
Ferguson 2011	Pressure support adjusted to ETT size, CPAP 5cmH_2O, $FiO_2 \leq 0.5$	2 hours
Loberger 2020	Pressure support 10cmH_2O regardless of ETT size, CPAP 5cmH_2O, FiO_2 0.5	2 hours

SBT = Spontaneous breathing trial, CPAP = Continuous positive airway pressure, ATC = Automated tube compensation, FiO_2 = Fraction of inspired oxygen, ETT = Endotracheal tube

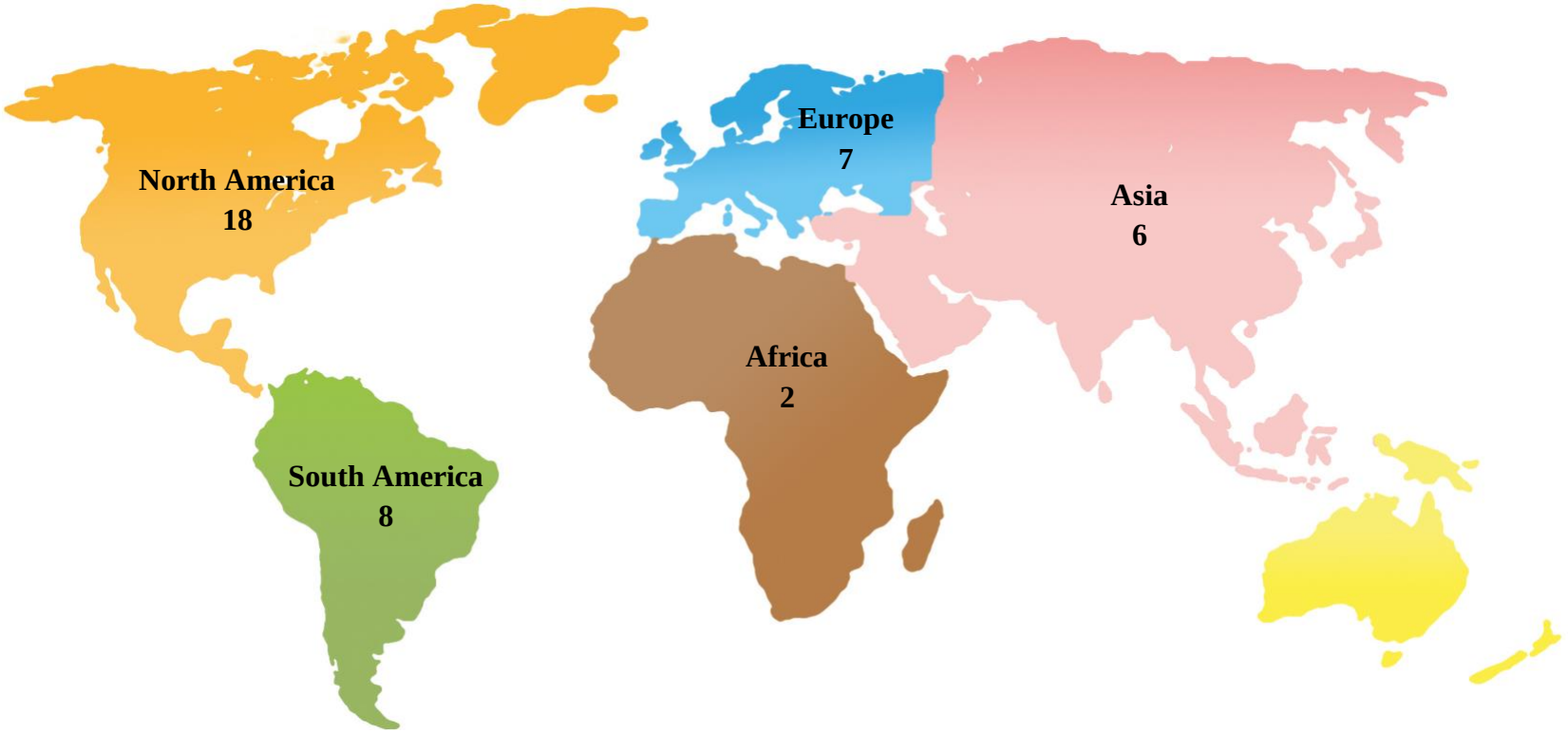


Figure S1 Representation of studies across the globe

References

1. Farias JA, Alía I, Retta A, Olazarri F, Fernández A, Esteban A, Palacios K, Di Nunzio L, Fernández G, Bordón A, Berrondo C, Sheehan G. An evaluation of extubation failure predictors in mechanically ventilated infants and children. *Intensive Care Med* 2002;28:752–757.
2. Farias JA., Alía I., Esteban A., Golubicki AN., Olazarri FA. Weaning from mechanical ventilation in pediatric intensive care patients. *Intensive Care Med* 1998;24:1070–5.
3. Manczur TI, Greenough A, Pryor D, Rafferty GF. Comparison of predictors of extubation from mechanical ventilation in children. *Pediatr Crit Care Med* 2000;1:28–32.
4. Thiagarajan RR., Bratton SL., Martin LD., Brogan TV., Taylor D. Predictors of successful extubation in children. *Am J Respir Crit Care Med* 1999;160:1562–6.
5. Harikumar G., Egberongbe Y., Nadel S., Wheatley E., Moxham J., Greenough A., Rafferty GF. Tension-time index as a predictor of extubation outcome in ventilated children. *Am J Respir Crit Care Med* 2009;180:982–8.
6. Khemani RG., Sekayan T., Hotz J., Flink RC., Rafferty GF., Iyer N., Newth CJL. Risk Factors for Pediatric Extubation Failure: The Importance of Respiratory Muscle Strength. *Crit Care Med* 2017;45:e798–e805.
7. Noizet O., Leclerc F., Sadik A., Grandbastien B., Riou Y., Dorkenoo A., Fourier C., Cremer R., Leteurtre S. Does taking endurance into account improve the prediction of weaning outcome in mechanically ventilated children? *Crit Care* 2005;9:R798-807.
8. Silva-Cruz A.L., Velarde-Jacay K., Carreazo N.Y., Escalante-Kanashiro R. Risk factors for extubation failure in the intensive care unit. *Rev Bras Ter Intensiva* 2018;30:294–300.
9. El-Beleidy ASE-D, Khattab AAE-H, El-Sherbini SA, Al-Gebaly HF. Automatic Tube Compensation versus Pressure Support Ventilation and Extubation Outcome in Children: A Randomized Controlled Study. *ISRN pediatrics* 2013;2013:871376.

10. Hubble CL., Gentile MA., Tripp DS., Craig DM., Meliones JN., Cheifetz IM. Deadspace to tidal volume ratio predicts successful extubation in infants and children. *Crit Care Med* 2000;28:2034–40.
11. Riou Y., Chaari W., Leteurtre S., Leclerc F. Predictive value of the physiological deadspace/tidal volume ratio in the weaning process of mechanical ventilation in children. *J Pediatr (Rio J)* 2012;88:217–21.
12. Laham JL., Breheny PJ., Rush A. Do clinical parameters predict first planned extubation outcome in the pediatric intensive care unit? *J Intensive Care Med* 2015;30:89–96.
13. Johnston C., de Carvalho WB., Piva J., Garcia PC., Fonseca MC. Risk factors for extubation failure in infants with severe acute bronchiolitis. *Respir Care* 2010;55:328–33.
14. Randolph AG., Forbes PW., Gedeit RG., Arnold JH., Wetzel RC., Luckett PM., O'Neil ME., Venkataraman ST., Meert KL., Cheifetz IM., Cox PN., Hanson JH., . Cumulative fluid intake minus output is not associated with ventilator weaning duration or extubation outcomes in children. *Pediatr Crit Care Med* 2005;6:642–7.
15. Fontela PS., Piva JP., Garcia PC., Bered PL., Zilles K. Risk factors for extubation failure in mechanically ventilated pediatric patients. *Pediatr Crit Care Med* 2005;6:166–70.
16. Toida C., Muguruma T., Miyamoto M. Detection and validation of predictors of successful extubation in critically ill children. *PloS one* 2017;12:e0189787.
17. Baumeister BL., el-Khatib M., Smith PG., Blumer JL. Evaluation of predictors of weaning from mechanical ventilation in pediatric patients. *Pediatr Pulmonol* 1997;24:344–52.
18. el-Khatib MF., Baumeister B., Smith PG., Chatburn RL., Blumer JL. Inspiratory pressure/maximal inspiratory pressure: does it predict successful extubation in critically ill infants and children? *Intensive Care Med* 1996;22:264–8.

19. Gehlbach JA, Miller AG, Hornik CP, Cheifetz IM. Dead Space to Tidal Volume Ratio Is Associated With Higher Postextubation Support in Children. *Respir Care* 2020;65:1721–1729.
20. Faustino EV., Gedeit R., Schwarz AJ., Asaro LA., Wypij D., Curley MA., . Accuracy of an Extubation Readiness Test in Predicting Successful Extubation in Children With Acute Respiratory Failure From Lower Respiratory Tract Disease. *Crit Care Med* 2017;45:94–102.
21. Loberger JM, Jones RM, Prabhakaran P. A Respiratory Therapist-Driven Pathway Improves Timeliness of Extubation Readiness Assessment in a Single PICU. *Pediatr Crit Care Med* 2020;21:e513–e521.
22. Chavez A., dela Cruz R., Zaritsky A. Spontaneous breathing trial predicts successful extubation in infants and children. *Pediatr Crit Care Med* 2006;7:324–8.
23. Farias JA., Retta A., Alía I., Olazarri F., Esteban A., Golubicki A., Allende D., Maliarchuk O., Peltzer C., Ratto ME., Zalazar R., Garea M., Moreno EG. A comparison of two methods to perform a breathing trial before extubation in pediatric intensive care patients. *Intensive Care Med* 2001;27:1649–54.
24. Foronda FK., Troster EJ., Farias JA., Barbas CS., Ferraro AA., Faria LS., Bousso A., Panico FF., Delgado AF. The impact of daily evaluation and spontaneous breathing test on the duration of pediatric mechanical ventilation: a randomized controlled trial. *Crit Care Med* 2011;39:2526–33.
25. Abu-Sultaneh S., Hole AJ., Tori AJ., Benneyworth BD., Lutfi R., Mastropietro CW. An Interprofessional Quality Improvement Initiative to Standardize Pediatric Extubation Readiness Assessment. *Pediatr Crit Care Med* 2017;18:e463–e471.

26. Nascimento MS., Rebello CM., Vale LAPA., Santos É., Prado CD. Spontaneous breathing test in the prediction of extubation failure in the pediatric population. *Einstein (Sao Paulo, Brazil)* 2017;15:162–166.
27. Keivanfar M, Sadeghizadeh A, Reisi M, Habibzadeh M, Babaei S. A Randomized Clinical Trial to Compare the Criteria of Readiness for Extubation and Daily Spontaneous Breathing Test (SBT) on the Duration of Mechanical Ventilation. *International Journal of Pediatrics* 2020;8:11003–11012.
28. Bilan N, Ganji S. Weaning from Ventilator and Effect of Blender-Humidifier on Outcome. *International Journal of Pediatrics* 2014;2:39–45.
29. Ferguson LP., Walsh BK., Munhall D., Arnold JH. A spontaneous breathing trial with pressure support overestimates readiness for extubation in children. *Pediatr Crit Care Med* 2011;12:e330-5.
30. Krasinkiewicz JM, Friedman ML, Slaven JE, Lutfi R, Abu-Sultaneh S, Tori AJ. Extubation Readiness Practices and Barriers to Extubation in Pediatric Subjects. *Respir Care* 2021;66:582–590.
31. Randolph AG., Wypij D., Venkataraman ST., Hanson JH., Gedeit RG., Meert KL., Luckett PM., Forbes P., Lilley M., Thompson J., Cheifetz IM., Hibberd P., Wetzel R., Cox PN., Arnold JH., . Effect of mechanical ventilator weaning protocols on respiratory outcomes in infants and children: a randomized controlled trial. *JAMA* 2002;288:2561–8.
32. Jouvet P., Farges C., Hatzakis G., Monir A., Lesage F., Dupic L., Brochard L., Hubert P. Weaning children from mechanical ventilation with a computer-driven system (closed-loop protocol): a pilot study. *Pediatr Crit Care Med* 2007;8:425–32.
33. Abdel Rahman DA, Saber S, El-Maghraby A. Diaphragm and Lung Ultrasound Indices in Prediction of Outcome of Weaning from Mechanical Ventilation in Pediatric Intensive Care Unit. *Indian J Pediatr* 2020;87:413–420.

34. IJland MM, Lemson J, van der Hoeven JG, Heunks LMA. The impact of critical illness on the expiratory muscles and the diaphragm assessed by ultrasound in mechanical ventilated children. *Ann Intensive Care* 2020;10:115.
35. Mistri S, Dhochak N, Jana M, Jat KR, Sankar J, Kabra SK, Lodha R. Diaphragmatic atrophy and dysfunction in critically ill mechanically ventilated children. *Pediatr Pulmonol* 2020;55:3457–3464.
36. Subhash S, Kumar V. Point-of-Care Ultrasound Measurement of Diaphragm Thickening Fraction as a Predictor of Successful Extubation in Critically Ill Children. *J Pediatr Intensive Care* 2021;doi:10.1055/s-0041-1730931.
37. Xue Y, Yang C-F, Ao Y, Qi J, Jia F-Y. A prospective observational study on critically ill children with diaphragmatic dysfunction: clinical outcomes and risk factors. *BMC Pediatr* 2020;20:422.
38. Valverde Montoro D, García Soler P, Hernández Yuste A, Camacho Alonso JM. Ultrasound assessment of ventilator-induced diaphragmatic dysfunction in mechanically ventilated pediatric patients. *Paediatr Respir Rev* 2021;S1526-0542(21)00005–1.doi:10.1016/j.prrv.2020.12.002.
39. Wratney AT., Benjamin DK., Slonim AD., He J., Hamel DS., Cheifetz IM. The endotracheal tube air leak test does not predict extubation outcome in critically ill pediatric patients. *Pediatr Crit Care Med* 2008;9:490–6.
40. Mhanna MJ., Zamel YB., Tichy CM., Super DM. The “air leak” test around the endotracheal tube, as a predictor of postextubation stridor, is age dependent in children. *Crit Care Med* 2002;30:2639–43.
41. Adderley RJ., Mullins GC. When to extubate the croup patient: the “leak” test. *Can J Anaesth* 1987;34:304–6.